

ES676 Membrane Processes

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Design of Drinking Water Treatment Plant Using Membrane Technology

Design of drinking water/industrial water treatment plant for a given forecasted year using membrane technology. The system is designed based on assumed data for per capita water supply, population forecasting (CPHEEO guidelines), and water quality parameters.

Given Data

- **Metropolitan city provided with piped water supply and existing sewerage system**
- **Base Year Population (2021) (assumed): 2200**
- **Forecast Year: 2030**
- **Growth Rate: 2% annually (as per CPHEEO geometric method)**
- **Source: Surface water**

Given Water Quality Parameters

Parameter	Value
Na ⁺	25 ppm
Mg ²⁺	40 ppm
Ca ²⁺	50 ppm
HCO ₃ ⁻	500 ppm
NO ₃ ⁻	10 ppm
Cl ⁻	30 ppm
SO ₄ ²⁻	5 ppm
Working Hours	12 hr/day

For a rapidly expanding metropolitan city, geometric increase method is used:

$$P_{2030} = P_{2021}(1 + 0.02)^9$$

$$P_{2030} = 2200(1.02)^9 \approx 2630$$

Based on CPHEEO for metropolitan city with piped water supply and existing sewerage system the water demand is 150 lpcd.

Average Daily Demand (ADD) = Population $\times$ Water Demand

$$ADD = 2630 \times 150 = 0.3945 \text{ MLD}$$

Maximum Daily Demand (MDD) = $1.8 \times ADD = 0.7101 \text{ MLD}$

Peak Hour Demand = $2.7 \times ADD = 1.065 \text{ MLD}$

Flow Rate, Q = $\frac{\text{MDD}}{\text{Working hours}}$

$$Q = \frac{0.7101 \times 10^6}{12 \times 1000} = 59.175 \text{ m}^3/\text{hr}$$

Considering Unaccounted Losses (15%) based on CPHEEO Recommendations:

$$Q = 1.15 \times 59.175 = 68.05 \text{ m}^3/\text{hr} \approx 68 \text{ m}^3/\text{hr}$$

Water Quality Parameters

$$\text{Total Hardness, } TH = \frac{50}{20} \times Ca^{2+} + \frac{50}{12} \times Mg^{2+}$$

$$TH = \frac{50}{20} \times 50 + \frac{50}{12} \times 40 = 291.67 \text{ mg/L as CaCO}_3$$

$$\text{Alkalinity} = \frac{500 \times 50}{61} = 409.8 \text{ mg/L as CaCO}_3$$

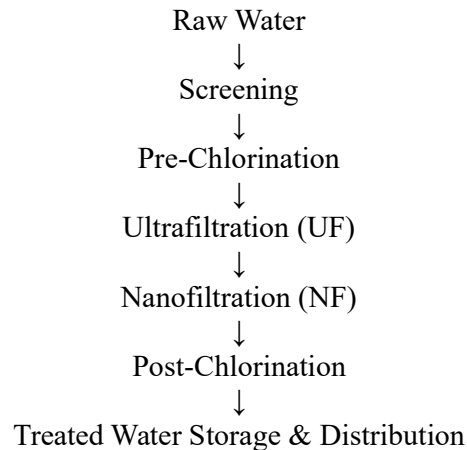
$$\text{TDS} = 25 + 40 + 50 + 500 + 30 + 5 + 10 = 660 \text{ mg/L}$$

For surface water with:

- Moderate TDS
- High hardness and alkalinity

Membrane-based treatment is selected.

Treatment Train:



Purpose of Each Unit

- **Screening:** Removal of large, suspended solids
- **Pre-chlorination:** Control of biological growth (excessive fouling)
- **UF:** Removal of turbidity, bacteria, viruses
- **NF:** Removal of hardness, alkalinity, dissolved salts
- **Post-chlorination:** Disinfection before distribution

Chemical Dosing

- **Antiscalant dosing:** 3 mg/L
 - Note: Antiscalant dosing is vendor-provided and depends on plant-specific scaling conditions. Detailed evaluation is required to finalize dosage.
- **Pre-chlorination:** 3 mg/L
- **Post-chlorination:** 2 mg/L

To supply 68 m³/hr of treated water, we must supply more feed water for treatment. Assume that UF Recovery is 90% and NF Recovery is 80%,

$$\text{Required Feed} = \frac{68}{0.90 \times 0.80} = \frac{68}{0.72} = 94.44 \approx 95 \text{ m}^3/\text{hr}$$

Design Parameters for UF:

- Design flux: 50 LMH (assumed)
- Membrane: Ultrafiltration IntegraTec XP 51

- Membrane area per module: 51 m² (From WAVE)

$$\text{Required Membrane Area} = \frac{95000}{50} = 1900 \text{ m}^2$$

$$\text{Number of Modules} = \frac{1900}{51} = 37.25$$

Using 36 modules in 2 treatment trains, there will be 18 modules per train and considering an additional train for standby provision, total modules = (2+1) × 18 = 54 modules

$$\text{Actual flux} = \frac{95000}{51 \times 36} = 51.74 \text{ LMH}$$

- One train can be taken offline for maintenance
- Continuous operation ensured
- Flux within acceptable operational limits

Design Parameter for NF:

- From UF permeate = 85.3 m³/hr
- Feed per pass = 68.3 m³/hr
- Membrane: NF 90-400/34i
- Area per element: 37.2 m²
- Elements per pressure vessel: 6
- Assumed Flux = 20 LMH

$$\text{Membrane Area Required} = \frac{68.3 \times 1000}{20} = 3415 \text{ m}^2$$

$$\text{Area per Vessel} = 37.2 \times 6 = 223.2 \text{ m}^2$$

$$\text{Number of Pressure Vessels} = \frac{3415}{223.2} \approx 15$$

Using two stage system for treatment in 2:1 array

- Stage 1 – 10 vessels
- Stage 2 – 5 vessels

Configuration works as:

- Feed enters Stage 1 (bulk removal)
- Concentrate flows to Stage 2 (polishing stage)
- Improves recovery and reduces wastage

Providing 1 standby unit for NF ensures:

- Plant remains operational during maintenance
- No interruption in treated water supply

The designed treatment plant uses UF and NF membrane processes to effectively remove turbidity, hardness, alkalinity, and dissolved solids. The modular design ensures operational flexibility, maintenance ease, and scalability for future demand.